Term 1: Heating and Cooling Checklist

Topic	Learning Outcomes	RAG Yourself	
		1	2
Heating and Cooling	Recall the three states of matter and the four changes of state		
	State the boiling and freezing point of water in degrees Celsius		
	Describe the how the particles are arranged differently in solids, liquids and gases		
	Explain the properties of solids, liquids and gases in terms of the arrangement of particles e.g. why can gasses be compressed?		
	Describe what is meant by expansion and contraction		
	Describe what happens to the particles in a substance when it is heated		
	Describe how thermal energy transfers between hotter and colder objects		
	Give the standard units of temperature and energy		
	Compare temperature to thermal energy in terms of kinetic energy of particles		
	Explain how colder objects can have more thermal energy than hotter objects		
	Describe how thermal energy transfer occurs through solids through conduction		
	Define the terms insulator and conductor with examples of each		
	Explain why metals are good conductors of energy		
	Describe how thermal energy transfers through liquids and gases by convection		
	Explain why the colour of a house is important when considering thermal insulation		